

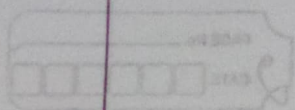
Q1) What is natural ecosystem?

A natural ecosystem, also known simply as an ecosystem, is a biological community of interacting organisms and their physical environment.

This community includes all living organisms (biotic factors) in a particular area, such as plants, animals, fungi, and microorganisms, as well as their non-living environment (abiotic factors) like soil, water, air, and sunlight.

Ecosystems vary widely in size and complexity, ranging from small ponds to vast rainforests. They can be terrestrial (land-based), aquatic (water-based), or a combination of both. 1

Natural ecosystems play crucial roles in supporting life on Earth by regulating important processes such as nutrient cycling, water purification, carbon sequestration, and climate regulation.



Q2) Why is a eco-system important?

Ecosystems are crucial for numerous reasons, both ecological and practical. Here are some key points highlighting their importance:

Biodiversity: Ecosystems support a wide variety of plants, animals, and microorganisms. This biodiversity is essential for the stability and resilience of ecosystems, as different species play unique roles in nutrient cycling, pollination, pest control, and other ecosystem functions.

Ecological Services: Ecosystems provide a multitude of services essential for human well-being, including clean air, and water, pollination of crops, soil fertility, climate regulation, and natural disaster mitigation (e.g., coastal protection provided by mangrove forests).

Economic Benefits: Ecosystems contribute directly and indirectly to economies through activities such as agriculture, fisheries, tourism, and pharmaceuticals. For e.g. - Forests provide timber, jobs, and medicinal resources, and tourism industries.

Soil conservation: Ecosystems help to maintain soil fertility and prevent erosion through the action of plant roots, ground cover, and organic matter decomposition.

Q3) Why should we conserve biodiversity?

Conserving biodiversity is crucial for a variety of reasons, encompassing ecological, economic, social, and ethical dimensions. Here are some key reasons why biodiversity conservation is important:

1. Ecosystem Stability and Resilience; Biodiversity contributes to the stability and resilience of ecosystems.

2. Ecosystem Services; Biodiversity underpins many ecosystem services essential for human well-being, including air and water purification, pollination of crops, soil fertility maintenance, climate regulation, and natural disaster mitigation.

medicinal Resources; many plant and animal species harbor valuable medicinal compounds used in traditional and modern medicine.

(Q4.) Desertification ⁱⁿ highly regions of the Himalayas?

Desertification in highly elevated regions to the Himalayas is not a common phenomenon. Desertification typically refers to the process of land degradation in arid, semi-arid, and dry sub-humid areas resulting from various factors such as climatic variations from various factors. Such as climatic variations and human activities.

Factors that could potentially lead to land degradation in the Himalayas or other mountainous regions include:

1. Deforestation: Clearing of forests for agriculture, logging or other purposes can lead to soil erosion and a loss of vegetation cover, making the land more susceptible to

degradation.

2. Overgrazing: Excessive grazing by livestock can degrade vegetation and contribute to soil erosion, particularly in alpine meadows.
3. Climate change: Change in climate patterns, including variations in precipitation and temperature, can impact the delicate ecosystems of the Himalayas and potentially contribute to land degradation.
4. Unsustainable agricultural practices: Poor agricultural practices, such as improper irrigation, excessive use of agrochemicals, and inadequate soil conservation measures, can contribute to soil degradation.

NA

Test-1

Ques

Q1. Environmental studies are

Ans (b) multidisciplinary field.

(2) The non-living components of an ecosystem are called.

Ans Abiotic factors

(3) What is the main source of energy for most ecosystems?

Ans ~~the~~ The Sun.

Q1. Differentiate b/w the Food chain and Food web.

Food chain:- A food chain is a linear flow of energy and nutrients from one organism to another.

Food web:- A food web can be termed as the combination of many different food chains and the relationship b/w organisms.

Food chain

Food web

- | | |
|--|--|
| <ul style="list-style-type: none"> A linear pathway showing the flow of energy | <ul style="list-style-type: none"> A multitude of networks showing the flow of energy |
| <ul style="list-style-type: none"> An organism of a high level trophic feeds on a specific organism or a lower trophic level. | <ul style="list-style-type: none"> An organism of a higher trophic level has access to more members of a lower trophic level. |

or) Give one example each of the Producer, Herbivore and Carnivore.

Producer :- Tree, plant.

eg of Carnivores are big cats such as tiger and cougar.

eg of herbivores are rabbit and grasshopper.

①

a) What is the importance of Public Awareness in environmental studies?

Public awareness and knowledge of environmental protection are crucial to avoid environmental pollutions.

b) Differentiate between renewable and non-renewable resources.

- Renewable resources cannot be depleted over time. Eg:- Water

- non-renewable resources deplete over time.

Eg:- Fossil fuel.

10/10

Signature

11. Living thing in an environment is known as _____
or _____.

(b) Biotic element.

12. The following is the correct food chain of grassland ecosystem.

(c) may be upright or inverted.

13. The pyramid of energy in an ecosystem.

(b) Grass → Grasshopper → Frog → Hawk → Snake.

Q14. _____ is the example of non-renewable natural resource.

(d) Coal and petroleum.

Q15. _____ is a fossil fuel.

(d) All the above.

16. The main purpose of rainwater harvesting is to recharge the _____.

(d) Groundwater table.

17 Biodiversity comprises

(a) All of the above

18 _____ indicator of Hot spot

(c) With high species richness of plants and high level of species endemism

19 Examples of in-situ conservation

(d) All of the above

20 Biosphere consists of

(a) All the above

21 ozone layer occurs in

(b) Stratosphere

22 Which of the following is a non-renewable energy resource?

(d) Coal

23 This is an example of non-polluting renewable type of energy.

14) All of these

24. this group consist of non-renewable organic resources

15) Natural gas, oil, and coal.

25. most of the energy used on farm today originally came from which of these sources?

16) The sun.

26. Which among the following is not a renewable source of energy?

17) Hydro-power.

27. The major non-renewable energy usage in India is _____

18) coal

28. Natural gas is a non-renewable energy

19) True.

20.

29. Which one of the following is a branch of forestry?

c) Forest management

30. The species which are in danger of extinction are called;

d) Endangered species